

DATA ANALYTICS CASE STUDY

AtliQ Hardware

Sales Insights & Profitability

A Power BI Business Analytics Project — End-to-End Data Analytics Case Study

This is a portfolio case study built on the publicly available AtliQ Hardware training dataset, using a self-directed, simulated stakeholder-feedback process.

₹985M TOTAL REVENUE <i>2017–2020, all markets</i>	~2M SALES QUANTITY <i>units sold, 2017–2020</i>	₹24.7M TOTAL PROFIT <i>aggregate, 2017–2020</i>	~150K TRANSACTIONS <i>sales_transactions records</i>
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Prepared by: Sahajahanur Rahman Laskar (Sahajan)

connectingsrl@gmail.com

linkedin.com/in/sahajahanur-laskar

github.com/Sahajahanur

Domain: Computer Hardware & Peripherals Distribution

Tools: MySQL | Power Query | Power BI | DAX

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1. Executive Summary

This report is a self-directed portfolio case study built on the publicly available AtliQ Hardware training dataset. AtliQ Hardware, its Sales Director, and the stakeholder feedback described in this report are part of the dataset's simulated business scenario, used here for learning and portfolio purposes — not an actual client engagement.

AtliQ Hardware is a computer hardware and peripherals distributor supplying retail clients such as Excel Stores across India. The Sales Director previously relied on verbal updates from Regional Managers and dozens of disconnected Excel files to gauge performance — a process that was slow, inconsistent, and prone to an overly optimistic ("rosy") bias, making it difficult to identify declining markets, underperforming products, or areas needing corrective action.

To solve this, this project built a governed, centralized Power BI analytics solution: sales data was extracted from the company's MySQL database, validated with SQL, cleaned and transformed in Power Query, modeled into a star schema, and enriched with a DAX measure library covering revenue, sales quantity, and profitability. The result is a three-page, scheduled-refresh Power BI dashboard that gives the Sales Director and Regional Managers a consistent, self-service view of performance by market, customer, and product.

The delivered analytical capabilities include revenue and sales-quantity tracking by market and customer, year-over-year trend comparison, Top 5 customer and product views, a Brick-and-Mortar vs. E-Commerce channel split, and — following stakeholder feedback — a full profitability layer (Profit Margin, Profit Margin %, Revenue Contribution %, and Profit Margin Contribution %) that was not part of the original scope.

Applying these capabilities to the dataset surfaced findings the business could not previously see: Delhi NCR generates 52.8% of total revenue but only 48.5% of profit contribution at a comparatively thin 2.3% margin; Mumbai is proportionally more profitable than its revenue share suggests; Bengaluru operates at a negative profit margin; and a single customer, Electricalsara Stores, accounts for roughly 42–46% of total revenue, indicating a customer-concentration risk alongside a multi-year revenue decline from 2017 to 2020.

2. Business Problem

2.1 Business Context

AtliQ Hardware supplies computer hardware and peripherals to retail clients across India, operating from a head office in Delhi with regional offices covering North, South, and Central India. Bhavan Patel, the company's Sales Director, is responsible for tracking performance across these regions.

2.2 Current Reporting Challenges

- The market is growing dynamically, and the Sales Director lacks a reliable, consistently updated way of tracking sales performance across regions.
- Insights are gathered by phone from Regional Managers, who tend to present an overly optimistic ("rosy") picture rather than the full truth.
- When hard numbers are requested, the business receives large volumes of disconnected Excel files that are too voluminous to manually consume or trust.
- There is no simple, digestible view showing which regions are declining and where corrective action (promotions, customer engagement) may be needed.

2.3 Business Questions

The dashboard was scoped to directly answer the following questions, each of which maps to a specific visual described in Section 8:

- How is revenue trending over time (2017–2020)?
- Which markets generate the most revenue, and which generate the most profit?
- Which customers contribute most to revenue, and which have weak margins?
- How does revenue differ between Brick-and-Mortar and E-Commerce channels?
- Where are underperforming markets or customers, relative to a configurable profit target?
- Is revenue concentration in a small number of customers creating business risk?

2.4 Desired Outcome

The desired outcome was a single, visual, self-service reporting tool that lets the Sales Director see the true state of the business at a glance, enabling data-driven rather than anecdote-driven decisions.

3. Project Scoping — AIMS Grid

Before any technical work began, the project was scoped using the AIMS Grid, a lightweight project-management framework covering Purpose, Stakeholders, End Result, and Success Criteria.

Component	Definition for this Project
Purpose	Solve the Sales Director's inability to get a clear, trustworthy, consistently updated picture of sales performance across regions, without relying on verbal reports or dozens of raw Excel files.
Stakeholders	Sales team, Marketing team, IT / software engineering team (owners of the sales management system), and the in-house Data Analytics team responsible for building the solution.
End Result	A Power BI dashboard, published to the Power BI Service with scheduled data refresh, giving the Sales Director and Regional Managers self-service access to revenue, sales quantity, and profitability trends.
Success Criterion (Target — not independently measured)	Reduce time spent on manual Excel consolidation by an estimated 10%, and improve sales visibility / decision speed by an estimated 5%, evaluated informally over a three-month period.

The success criteria above were defined as project targets at the scoping stage. This project did not include a formal, measured before/after study of cost savings or decision-speed improvement; Section 14 presents these as expected — not achieved — business impact.

This structured scoping step ensured the project had a clearly defined owner, audience, and measurable definition of "done" before any data work began — mirroring how analytics projects are typically initiated in a corporate environment.

4. Data Discovery & SQL Analysis

4.1 Source System

AtliQ Hardware's sales data lives in a MySQL database owned by the company's software engineering team (referred to internally as the "Falcons"), who maintain the Sales Management System that generates an invoice for every transaction. The Data Analytics team was granted read access to this database for reporting purposes.

In organizations with high transaction volume, direct analytical querying against the production OLTP (Online Transaction Processing) system is generally avoided, since heavy analytical queries can degrade the performance of a mission-critical transactional system; the standard pattern is to Extract, Transform, and Load (ETL) data into a separate analytical store. For the scale of this dataset (~150,000 records), the MySQL database was connected to directly within Power BI via Power Query, and all cleaning/transformation was performed there rather than in a dedicated data warehouse — an appropriately simple architecture for the data volume involved.

4.2 Source Tables

Table	Type	Approx. Records	Key Fields
sales_transactions	Fact	~150,000	product_code, customer_code, market_code, order_date, sales_qty, sales_amount, currency
customers	Dimension	38	customer_code, customer_name, customer_type
products	Dimension	—	product_code, product_name
markets	Dimension	—	market_code, markets_name, zone
date	Dimension	—	date, cy_date (month-start), year

4.3 Exploratory Analysis

Before building any dashboard, exploratory SQL queries were run in MySQL Workbench to understand data volume, validate figures, and surface anomalies. Representative queries included:

- Row-count checks — e.g., `SELECT COUNT(*) FROM sales_transactions;` to confirm total transaction volume.
- Market-level filtering — e.g., `SELECT * FROM sales_transactions WHERE market_code = 'Mark001';` to isolate a single city's transactions.
- Year-over-year revenue validation — e.g., `SELECT SUM(sales_amount) FROM sales_transactions st JOIN date d ON st.order_date = d.date WHERE d.year = 2020;` repeated for each year of interest.
- Currency distribution checks — e.g., `SELECT DISTINCT currency FROM sales_transactions;` used to detect unexpected duplicate currency labels.

4.4 Data Quality Findings

The exploratory queries surfaced several data-quality issues typical of a long-lived production database — non-India market records, invalid (negative or zero) sale amounts, mixed currencies, and duplicated currency labels caused by a stray control character. These findings directly shaped the Power Query cleaning steps documented in Section 5.

4.5 Validation Results

SQL-based aggregation was used as an independent check against the figures later produced in Power BI. Direct SQL queries against the source data returned a 2020 total revenue of ₹142.22M and a 2019 total revenue of ₹336.02M —

an early, SQL-verified signal of the year-over-year revenue decline that motivated the project, and a benchmark later used to confirm that the Power BI measures were calculating correctly.

5. Data Cleaning & ETL (Power Query)

Data quality issues identified during SQL exploration were resolved in Power Query. Each transformation step was named descriptively so the pipeline reads as a clear, auditable sequence — distinguishing what was a data issue in the source, what was resolved within the ETL pipeline, and what was logged back to the engineering team rather than silently altered.

5.1 Data Quality Issues & Resolutions

Data Quality Issue	Type	Resolution
Non-India markets (New York, Paris) with a blank zone	Source data issue	Filtered out rows where zone was blank
Negative and zero values in sales_amount	Source data issue	Filtered out in Power Query; also reported to the engineering team as a likely source-system defect
Mixed currencies (INR and USD) in one amount column	Source data issue / business rule needed	Resolved in ETL — added a Normalized Sales Amount column converting USD to INR
Duplicate currency labels: "INR"/"INR\r", "USD"/"USD\r"	Source data issue (stray control character)	Standardized in ETL; duplicate-currency records investigated and removed after confirming they were duplicate transactions
Normalized Sales Amount loaded as Text instead of Decimal	ETL configuration issue	Corrected in Power Query — converted the column's data type to Decimal Number so it would aggregate correctly
Blank product_code in some records	Source data issue	Documented as a known limitation and routed back to the engineering team rather than silently dropped (see Section 13)

5.2 ETL Pipeline (Power Query Steps)

1. Source — Connect to MySQL, load all five tables.
2. Filter Markets — Remove New York/Paris (blank-zone) records from the markets table.
3. Remove -1 and 0 in Sales — Filter out invalid sales_amount values.
4. Clean Up Currency — Standardize currency codes and remove the duplicate \r-suffixed variants.
5. Add Normalized Sales Amount Column — Conditional column converting USD to INR.
6. Fix Data Type — Corrected Normalized Sales Amount from Text to Decimal Number.
7. Close & Apply — Load the cleaned model into the Power BI data model.

5.3 Currency Normalization Assumption

A small number of transactions in the source data were recorded in USD rather than INR. To enable accurate aggregation, these were converted to INR in Power Query using a single, fixed conversion rate documented in the project notes. This was a deliberate simplification for the scope of this project.

Limitation: a production implementation should replace the fixed conversion rate with an effective-date exchange-rate table (converting each transaction at the rate in effect on its transaction date), since currency rates move over time and a static multiplier introduces a small, systematic distortion for older transactions.

6. Data Modeling — Star Schema

The five tables were connected in a star schema — the standard data-modeling pattern for BI reporting, with one central fact table surrounded by descriptive dimension tables. A star schema was appropriate here because it keeps relationships simple (single-direction, one-to-many), performs well for the aggregation-heavy queries a sales dashboard requires, and is easy for a report author to reason about — every dimension joins directly to the one fact table rather than through intermediate tables.

Dimension	Fact	Key	Cardinality	Filter Direction
Customers	Sales Transactions	customer_code	1 : many	Single
Products	Sales Transactions	product_code	1 : many	Single
Markets	Sales Transactions	market_code	1 : many	Single
Date	Sales Transactions	date ↔ order_date	1 : many	Single

The Customers and Products relationships were auto-detected by Power BI on load, since the join-key field names matched exactly. The Markets and Date relationships required manual creation, because the source field names did not match exactly: the Markets dimension's market_code needed to be explicitly linked to the fact table's market_code, and the Date dimension's date field was related to the fact table's order_date field — a common real-world data-modeling step rather than a Power BI default.

7. DAX Measures & Calculation Architecture

A dedicated Base Measures table was created to hold every calculated metric, separating calculation logic from raw data columns. This keeps the model organized, makes measures easy to find and reuse across all three dashboard pages, and avoids scattering DAX logic inside individual visuals.

Measure	Business Purpose	Logic (plain-language)
Revenue	Total sales value	SUM of Normalized Sales Amount
Sales Quantity	Total units sold	SUM of sales_qty
Profit Margin	Total profit generated	SUM of the transaction-level profit margin field
Profit Margin %	Profitability relative to revenue, for the current filter context (market / product / customer)	Profit Margin ÷ Revenue, using DIVIDE() to safely return 0 rather than an error when Revenue is 0
Revenue Contribution %	Share of total company revenue contributed by the current market or customer	Current-context Revenue ÷ Revenue calculated across ALL markets, products, and customers

Measure	Business Purpose	Logic (plain-language)
Profit Margin Contribution %	Share of total company profit contributed by the current market or customer	Current-context Profit Margin ÷ Profit Margin calculated across ALL markets, products, and customers

The two "contribution" measures were the key technical addition in the project's second phase. Filter-context measures (Revenue, Profit Margin) naturally answer "how big is this market?" within whatever slice is currently selected, but they cannot on their own answer "how much of the total does this market represent?" — that requires deliberately overriding the current filter context using ALL() to compute a company-wide total, then dividing the local value by it. DIVIDE() was used throughout rather than the standard division operator, since it returns a safe default (0) instead of an error when the denominator is 0 — important for markets or customers with no recorded revenue in a given period. Together, these measures let the same visual answer both "how big is this market?" (revenue) and "how valuable is this market?" (profit contribution), which turned out to tell two different stories for the business (see Section 11).

8. Dashboard Design & Development

The dashboard was built iteratively across three pages, moving from a first working version, to a profitability-focused view added after stakeholder feedback, to a consolidated performance-monitoring view.

Page 1 — Sales Overview

Purpose: give a high-level view of revenue and sales performance.

Includes: Revenue and Sales Quantity KPI cards; Revenue and Sales Quantity by Market; a Revenue Trend line; Top 5 Customers by Revenue; Top 5 Products by Revenue; a Brick-and-Mortar vs. E-Commerce channel split; and Year and Month slicers.

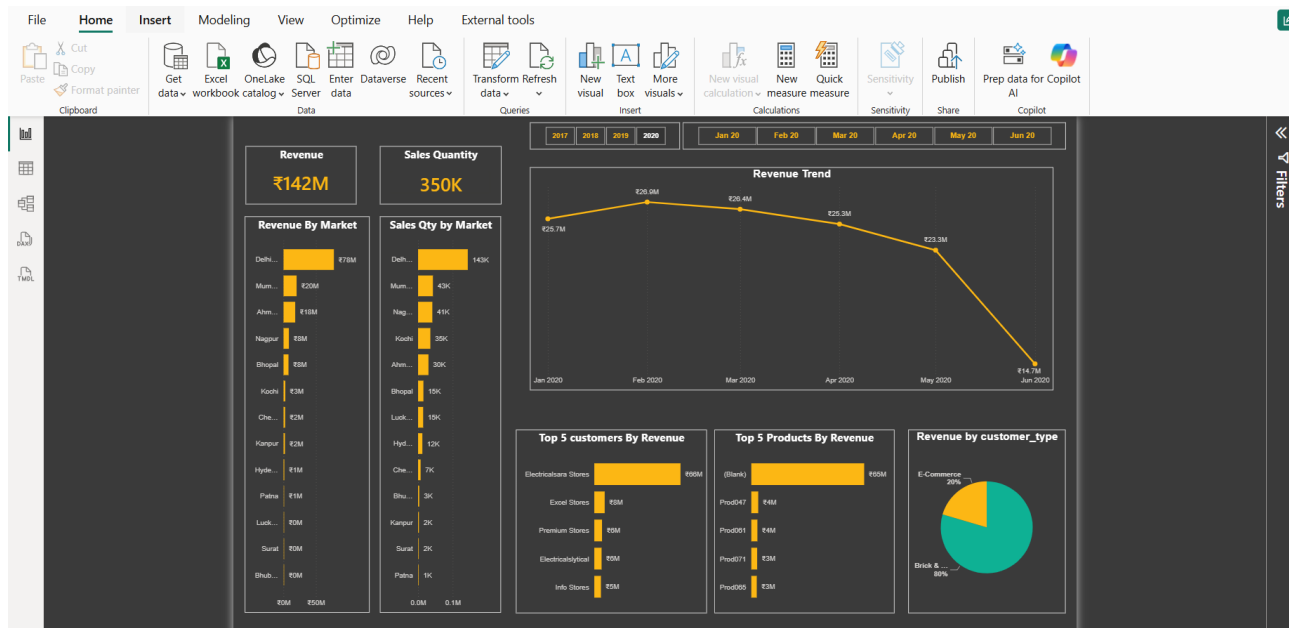


Figure 1 — Page 1 (Sales Overview): Revenue, Sales Quantity, Revenue/Sales Qty by Market, Revenue Trend, Top 5 Customers, Top 5 Products, and channel split, filtered to 2020.

Page 2 — Profitability & Customer Performance

Purpose: understand profitability rather than revenue alone.

Following stakeholder feedback (Section 9), this page was added to answer a question Page 1 could not: which markets and customers are actually profitable, not just which generate the most revenue. Includes: Revenue Contribution % by Market, Profit Contribution % by Market, Profit % by Market, a Revenue Trend chart, and a full Customer Performance Summary table.

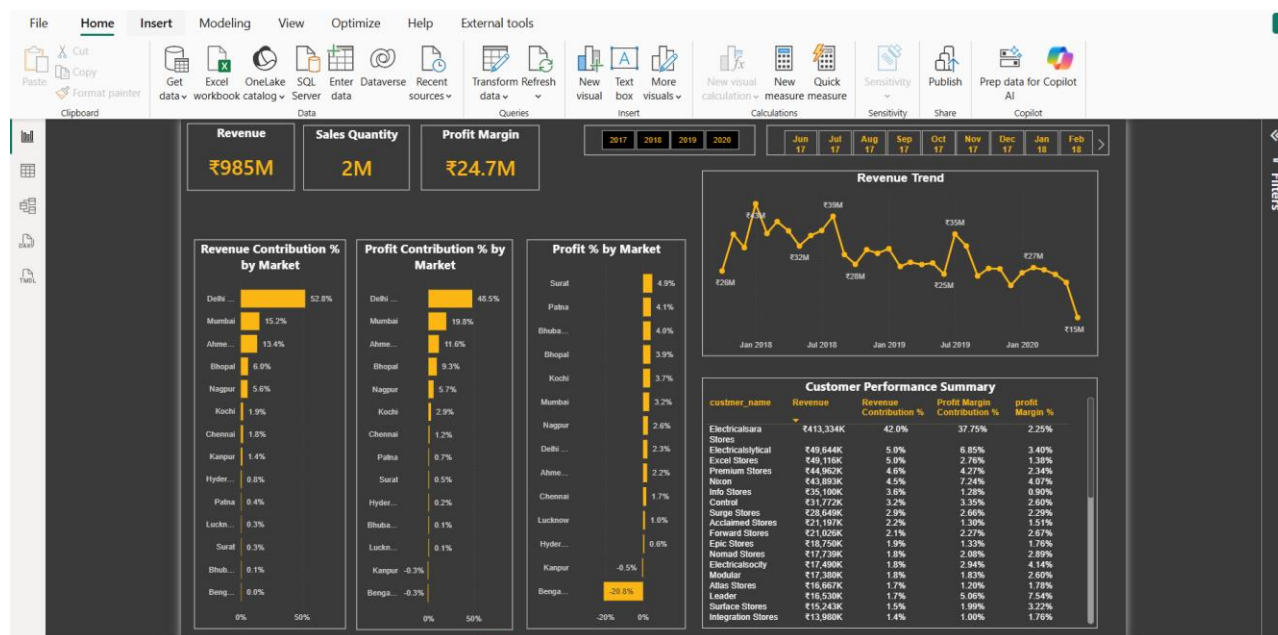


Figure 2 — Page 2 (Profitability & Customer Performance): Revenue Contribution % by Market, Profit Contribution % by Market, Profit % by Market, Revenue Trend, and Customer Performance Summary table.

Page 3 — Consolidated Performance

Purpose: combine key performance tracking with exception monitoring in a single view.

Includes: Revenue Contribution % by Market, a configurable Profit Target parameter, a Revenue Trend chart overlaying current-year Revenue, prior-year Revenue, and Profit Margin %, and a sortable Customer Performance Summary matrix (Revenue, Revenue Contribution %, Profit Margin Contribution %, and Profit Margin % side by side per customer).

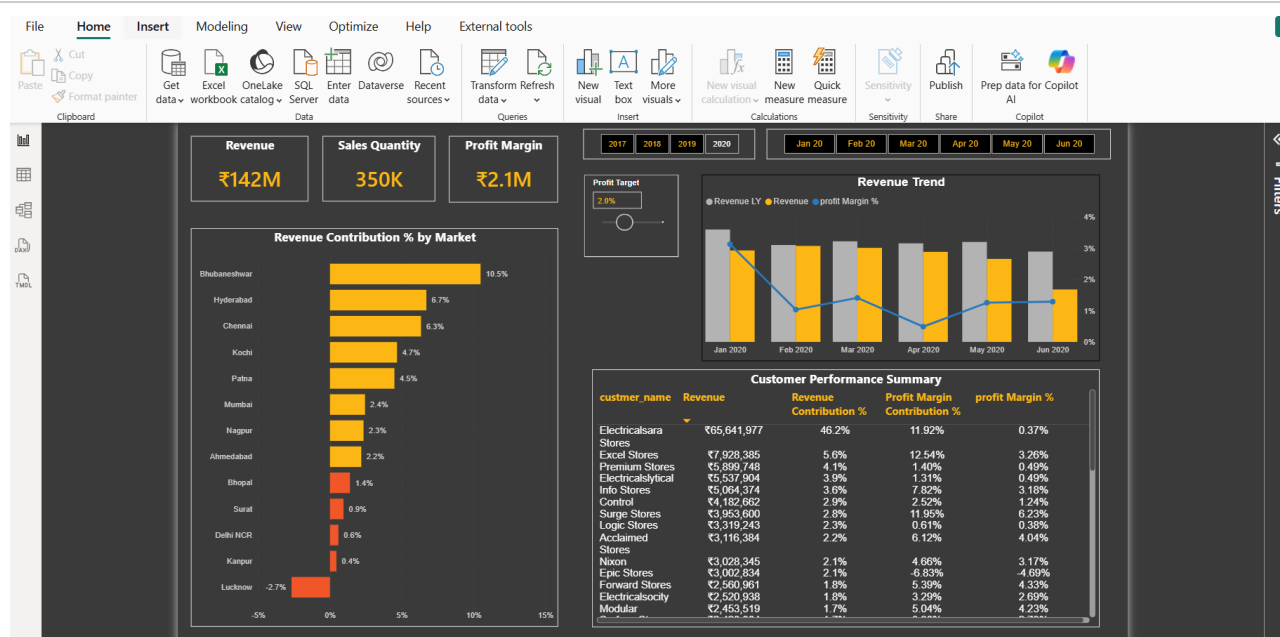


Figure 3 — Page 3 (Consolidated Performance): Revenue Contribution % by Market with a Profit Target slider, Revenue Trend overlay, and full customer-level profitability matrix.

Formatting & UX Decisions

- Cross-filtering (instead of the default cross-highlighting) was enabled globally under File → Options → Report Settings, so clicking one visual filters related visuals to only relevant data rather than dimming irrelevant bars.
- Large numbers were displayed in millions/thousands (e.g., ₹142M rather than ₹142,220,000) for at-a-glance readability.
- Conditional formatting, driven by the Profit Target parameter, visually flags markets performing below a configurable profit-margin threshold, turning the chart into an exception-reporting tool rather than a static ranking.
- A dedicated Mobile Layout was built in Power BI Desktop (View → Mobile Layout) so the dashboard renders correctly in the Power BI mobile app.

9. Stakeholder Feedback & Iteration

After the first version of the dashboard (Page 1) was built, it was shared with a simulated stakeholder review — reflecting how a dashboard evolves after being handed to business users in a real organization. This exercise was conducted as part of the project's self-directed learning process, not a live corporate deployment; the feedback items below reflect that structured review rather than an external client engagement.

Stakeholder Question / Feedback	Analytical Requirement	Dashboard Change / Business Value
Revenue measure was still using raw Sales Amount instead of the currency-normalized amount.	Correct the Revenue measure's source column.	Bug fixed — Revenue measure repointed to Normalized Sales Amount; a related data-type issue (text vs. decimal) was caught and corrected in the same pass.
"Add product cost and margin data — revenue alone doesn't show what's actually working."	Introduce profitability, not just revenue, into the model.	Added Cost Price and Profit Margin fields and built the full profitability measure set described in Section 7.

Stakeholder Question / Feedback	Analytical Requirement	Dashboard Change / Business Value
"Which markets/customers give the best return, not just the most revenue?"	Compare markets on a profit basis, not only a revenue basis.	Built Profit Margin %, Profit Margin Contribution %, and Revenue Contribution % visuals; revealed that Mumbai is proportionally more profitable than Delhi NCR despite lower revenue.
"Show a two-dimensional customer view — margin % and contribution together."	One table combining revenue scale and profitability.	Built the tabular Customer Performance Summary combining Revenue, Revenue Contribution %, Profit Margin Contribution %, and Profit Margin % per customer.
"Split revenue by Brick-and-Mortar vs. E-Commerce."	Channel-level revenue visibility.	Added a customer-type pie chart to Page 1 (Section 8).
"Break performance down by zone, and flag underperforming markets."	Threshold-based exception flagging.	Added a dynamic Profit Target parameter with conditional formatting to flag markets below target on Page 3.
"Show number of orders, not just quantity, to understand fulfillment/shipping cost."	Order-count-level granularity.	Identified as a data-availability gap — the source system captures line-level quantity, not a distinct order-count field; logged as a data request to the engineering team, not implemented in this project.
"Only 3–4 years of history — can we see more?"	Longer historical trend.	Logged as a data-availability limitation, not a dashboard defect — historical data beyond what exists in the source system cannot be added without a data-engineering change.
"Can decisions/comments be recorded against the data?"	Write-back / annotation capability.	Assessed as technically feasible via the Microsoft Power Platform (e.g., Power Apps with a SQL-backed table), but explicitly out of scope for this project — documented in Section 12 as a Future Enhancement, not implemented.

This iteration cycle is documented deliberately: it demonstrates how a dashboard evolves as a living product shaped by stakeholder feedback, rather than a one-time deliverable, and shows the analyst's process of triaging feedback into fixes, new features, and logged data-availability gaps.

10. Final Dashboard Walkthrough

The completed three-page dashboard delivers:

- Headline KPI cards — Revenue, Sales Quantity, and Profit Margin, refreshed from the source system on a scheduled basis.
- Year and month slicers for flexible, interactive time-based filtering.
- Market-level revenue, sales quantity, revenue-contribution %, profit-contribution %, and profit % views.

- A Revenue Trend chart overlaying current revenue, prior-year revenue, and profit margin % for year-over-year comparison.
- A full Customer Performance Summary table enabling multi-dimensional (revenue × margin × contribution) analysis per customer.
- Top 5 Customers and Top 5 Products views.
- A Brick-and-Mortar vs. E-Commerce revenue split.
- A configurable Profit Target parameter with conditional formatting to flag under-target markets.

Live report: [Live Power BI Report](#) (sign-in with an authorized account required).

11. Key Business Insights

11.1 Revenue vs. Profit Are Not Aligned

Delhi NCR contributes 52.8% of total revenue but only 48.5% of total profit contribution, at a comparatively modest 2.3% profit margin — the highest-revenue market is not the highest-return market. The margin pattern warrants further investigation into pricing, discounting, product mix, and other cost drivers before any conclusion about root cause is drawn.

11.2 Mumbai Shows Stronger Relative Profitability

Despite ranking second in revenue (15.2% of the total), Mumbai contributes a disproportionately high share of total profit relative to its revenue share. This pattern is worth evaluating further — possible contributing factors could include pricing, discounting practices, or product mix, but the dashboard alone does not isolate the cause.

11.3 One Market Shows a Negative Profit Margin

Bengaluru shows a –20.8% profit margin on a very small revenue share. This is a clear candidate for further review of pricing, discount policy, and product mix in that market; the data indicates a pattern worth investigating rather than a confirmed root cause.

11.4 Revenue Concentration in a Small Customer Base

Electricalsara Stores alone accounts for roughly 42–46% of total revenue depending on the period analyzed, at a comparatively thin profit margin (approximately 0.4%–2.3%) — a high absolute profit contribution driven primarily by transaction volume rather than margin. This concentration represents both a growth engine and a customer-concentration risk that should be assessed as part of broader account-management strategy.

11.5 Declining Revenue Trend

The historical trend across the 2017–2020 analysis period shows overall revenue trending down from an earlier peak toward 2020, consistent with the Sales Director's original concern. This reinforces the value of the market- and customer-level profitability views added in this project, so that any recovery strategy can be targeted rather than broad-brush; the dashboard highlights the pattern but does not itself diagnose the underlying cause.

11.6 Channel Mix

For the 2020 filter view, the business remains predominantly Brick-and-Mortar (approximately 80%) versus E-Commerce (approximately 20%), giving a clear baseline against which future digital-channel initiatives could be measured.

12. Publishing & Deployment

Implemented

- Published from Power BI Desktop to the Power BI Service (cloud), accessible to any authorized user via a work-email login, without requiring Power BI Desktop installed locally.
- Scheduled Refresh — configured so the dataset can be refreshed automatically on a defined schedule (e.g., a few times per day) rather than requiring manual updates. This is a scheduled-refresh reporting layer, not a real-time streaming pipeline — the data is only as current as the most recent scheduled refresh.
- Personal & Report Bookmarks — e.g., saved views for Delhi, Mumbai, and Ahmedabad, enabling one-click, presentation-style walkthroughs without a separate slide deck.
- Comments — enabling threaded, in-context discussion between stakeholders and the analyst (the mechanism used for the feedback cycle in Section 9).
- Export to PowerPoint / PDF — for use in executive presentations.
- Mobile Layout — a custom mobile view published so users can view KPIs, trends, and top-customer/product views from the Power BI mobile app.

Future Enhancement (Not Implemented in This Project)

- Write-back / comment-to-data capability via the Microsoft Power Platform (e.g., Power Apps with a SQL-backed comments table), so decisions made in review meetings could be recorded directly against the data — assessed as feasible but out of scope here.
- A dedicated data warehouse layer (e.g., Snowflake) and an effective-date currency exchange-rate table, for a larger-scale or production deployment (see Section 13).
- Distinct order-count capture at the source-system level, to support fulfillment/shipping-cost analysis.

13. Limitations & Assumptions

The following are known limitations and opportunities for future iteration, rather than defects in the delivered solution:

- Historical data availability: the source system contains approximately four calendar years of transaction history (2017–2020); trend analysis cannot extend further back without a source-system change.
- No distinct order-count field: the source data captures line-level sales quantity, not a separate order identifier, so order-level metrics (e.g., average order value, shipping-cost analysis) are not currently possible.
- Source-system data-quality issues: negative/zero sale amounts, blank product mappings, and duplicate currency labels were observed in the raw data and were filtered, corrected, or logged back to the engineering team as documented in Section 5 — these reflect the source system's data quality, not the analysis.
- Simplified currency conversion: USD-to-INR conversion used a single fixed rate rather than an effective-date exchange-rate table (Section 5.3).
- Scheduled refresh, not real-time: the dashboard reflects data as of its most recent scheduled refresh, not live/streaming data.

- Simulated stakeholder environment: the feedback cycle in Section 9 was conducted as a structured, self-directed review exercise rather than a live engagement with AtliQ Hardware's actual business stakeholders.
- Success criteria not independently measured: the AIMS Grid success criteria (Section 3) were defined as project targets; this project did not include a formal before/after measurement of cost savings or decision-speed improvement.

14. Recommendations & Expected Business Impact

Recommendations

Insight	Recommendation
Bengaluru shows a negative profit margin.	Investigate pricing, discounting, and product mix in Bengaluru before allocating further growth budget to that market.
Delhi NCR generates the most revenue but a comparatively thin margin.	Evaluate Delhi NCR's cost structure and discounting practices relative to margin, rather than judging the market on revenue alone.
Mumbai shows stronger relative profitability than its revenue share suggests.	Study Mumbai's pricing, discounting, and product-mix practices as a potential reference point for other regions.
Revenue is heavily concentrated in one customer (Electricalsara Stores).	Develop the next tier of customers to reduce concentration risk, while continuing to protect the relationship that currently anchors a large share of total revenue.
Revenue has trended down over 2017–2020.	Use the market- and customer-level profitability views to target recovery efforts rather than applying broad, undifferentiated promotions.
Channel mix is heavily Brick-and-Mortar (~80%).	Monitor E-Commerce growth over time as a potential diversification opportunity.

Expected Business Impact

These outcomes were not independently measured during this project and are presented as expected, not achieved, impact:

- Reduced manual reporting effort — a self-service dashboard is designed to reduce time spent compiling and reconciling Excel files across regions.
- Faster access to information — scheduled refresh and mobile access could reduce the delay between a change in business performance and its visibility to decision-makers, compared with phone-based reporting.
- Improved profitability visibility — the Profit Margin, Profit Margin %, and Contribution % measures give the business a profitability lens that did not exist in the original Excel-based process.
- Easier exception identification — the Profit Target conditional formatting is designed to make under-target markets easier to spot than scanning raw tables.
- Improved stakeholder self-service — bookmarks, mobile layout, and comments are designed to reduce the Sales Director's dependence on ad hoc requests to the analytics team.

15. Conclusion — Final Project Outcome

What Was Built

A three-page Power BI dashboard, backed by a cleaned MySQL data source, a star-schema data model, and a DAX measure library, published to the Power BI Service with scheduled refresh and a dedicated mobile layout.

What Problems It Solved

Replaced a fragmented, verbally-reported, Excel-heavy sales reporting process with a centralized, self-service reporting layer covering revenue, sales quantity, and — after stakeholder feedback — profitability.

What Analytical Capabilities Were Added

Market- and customer-level revenue and profit analysis, contribution-percentage measures, year-over-year trend comparison, channel-mix analysis, and threshold-based exception flagging via a configurable Profit Target.

What Business Insights Were Discovered

Revenue and profitability are not aligned across markets; a small number of customers account for a large share of revenue; one market operates at a loss; and overall revenue has declined across the analysis period — findings that were not visible under the previous, manual reporting process.

What Limitations Remain

Order-count granularity, longer historical depth, a production-grade currency-conversion approach, and independently measured business impact remain open items for future iterations (Section 13).

This project was delivered through the same lifecycle used in professional analytics teams: problem definition, AIMS-grid scoping, SQL-based data discovery, Power Query ETL, star-schema data modeling, DAX measure development, iterative dashboard design, stakeholder-feedback-driven refinement, and cloud publishing — demonstrating an end-to-end Data Analyst / BI Analyst workflow from a raw operational database to a decision-ready reporting product.

16. Appendix — Technical Stack

Category	Tools / Techniques Used
Database	MySQL, MySQL Workbench
SQL	SELECT, JOIN, GROUP BY, aggregation, filtering — used for exploratory analysis and validation
ETL	Power Query (M) — filtering, conditional columns, currency normalization, data-type correction
Data Modeling	Star schema (1 fact table, 4 dimension tables)
BI Platform	Power BI Desktop
Calculations	DAX — SUM, DIVIDE, ALL, iterative contribution measures
Deployment	Power BI Service (cloud publishing, scheduled refresh)
UX / Collaboration	Slicers, bookmarks, conditional formatting, comments, mobile layout
Project Method	AIMS Grid scoping; stakeholder-feedback-driven iteration

End of report.